

Conversion Log

Production Conversion Log

March 2025 · Girardot, Colombia

Submission reference	RTF0-310825
Reporting period	March 2025 (2025-03)
Plant	Girardot, Colombia
Product	DEV-P100 — refined pyrolysis oil
Off-taker	Crown Oil Limited (UK)
Regulator	UK DfT — LCF Delivery Unit
Density — EU-grade	0,782 kg/L
Density — PLUS-grade	0,858 kg/L
Production days observed	24 of 31 (7 non-production days)
Source	daily_production table (DFT backend, snapshot 2026-05-21T00:00:00Z)
Generated	2026-05-21T00:00:00Z

Signed for and on behalf of OisteBio GmbH

Paolo Ughetti

Managing Director

Date: _____

Executive Summary

Per-day kilogram-to-litre conversion log for **March 2025** production at the Girardot, Colombia pyrolysis facility, operated by **OisteBio GmbH**. Output product is **DEV-P100 — refined pyrolysis oil**, supplied to **Crown Oil Limited (UK)** under the UK DfT — LCF Delivery Unit reporting framework. Litres are derived from kilograms via the nominal product densities listed below, as materialised by the `litres_eu` and `litres_plus` GENERATED ALWAYS columns on the `daily_production` table (Alembic migration `0007_persist_production_litres.py`).

Monthly aggregate

Metric	Value
Feedstock routed to thermal conversion (input)	2 886 397,000 kg
Total EU-grade production	744 690,426 kg / 952 289,547 L
Total PLUS-grade production	1 227 911,768 kg / 1 431 132,598 L
Liquid-product yield (EU + PLUS) over input	68,34 %
Grand total volume (EU + PLUS)	2 383 422,145 L

Co-products tracked separately on the same daily ledger (not converted to litres because they are solid / gaseous): carbon black **399 785,226 kg**, steel scrap **268 427,226 kg**, water condensate **48 545,350 kg**, syngas **135 641,417 kg**, recorded losses **61 395,587 kg**. Mass-balance closure across all streams is reported in Annex A.

Daily Conversion — March 2025

One row per production day from the `daily_production` table. Days with no production record (weekends, public holidays, maintenance) are listed in the gap line below. Litres computed at constant density: **EU 0,782 kg/L, PLUS 0,858 kg/L**. Liquid yield = (EU kg + PLUS kg) / input kg.

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-03-01	129 238,000	33 343,404	54 979,568	0,782	0,858	42 638,624	64 078,751	68,34 %	—
2025-03-03	106 135,000	27 382,830	45 151,244	0,782	0,858	35 016,407	52 623,828	68,34 %	—
2025-03-04	118 817,000	30 654,786	50 546,337	0,782	0,858	39 200,494	58 911,815	68,34 %	—
2025-03-05	136 276,000	35 159,208	57 973,627	0,782	0,858	44 960,624	67 568,330	68,34 %	—
Total	2 886 397,000	744 690,426	1 227 911,768	—	—	952 289,547	1 431 132,598	68,34 %	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-03-06	121 889,000	31 447,362	51 853,206	0,782	0,858	40 214,018	60 434,972	68,34 %	—
2025-03-07	143 377,000	36 991,266	60 994,487	0,782	0,858	47 303,409	71 089,146	68,34 %	—
2025-03-08	144 606,000	37 308,348	61 517,320	0,782	0,858	47 708,885	71 698,508	68,34 %	—
2025-03-10	131 803,000	34 005,174	56 070,754	0,782	0,858	43 484,877	65 350,529	68,34 %	—
2025-03-11	127 242,000	32 828,436	54 130,443	0,782	0,858	41 980,097	63 089,094	68,34 %	—
2025-03-12	121 394,000	31 319,652	51 642,626	0,782	0,858	40 050,706	60 189,541	68,34 %	—
2025-03-13	95 738,000	24 700,404	40 728,222	0,782	0,858	31 586,194	47 468,790	68,34 %	—
2025-03-14	129 101,000	33 308,058	54 921,287	0,782	0,858	42 593,425	64 010,824	68,34 %	—
2025-03-15	88 298,000	22 780,884	37 563,147	0,782	0,858	29 131,565	43 779,892	68,34 %	—
2025-03-17	127 232,000	32 825,856	54 126,189	0,782	0,858	41 976,798	63 084,136	68,34 %	—
2025-03-18	110 954,000	28 626,132	47 201,311	0,782	0,858	36 606,307	55 013,183	68,34 %	—
2025-03-19	133 583,000	34 464,414	56 827,989	0,782	0,858	44 072,141	66 233,087	68,34 %	—
2025-03-20	137 550,000	35 487,900	58 515,604	0,782	0,858	45 380,946	68 200,005	68,34 %	—
2025-03-21	124 994,000	32 248,452	53 174,114	0,782	0,858	41 238,430	61 974,492	68,34 %	—
2025-03-22	96 020,000	24 773,160	40 848,188	0,782	0,858	31 679,233	47 608,611	68,34 %	—
2025-03-25	123 410,000	31 839,780	52 500,259	0,782	0,858	40 715,831	61 189,113	68,34 %	—
2025-03-26	123 043,000	31 745,094	52 344,133	0,782	0,858	40 594,749	61 007,148	68,34 %	—
2025-03-27	105 218,000	27 146,244	44 761,140	0,782	0,858	34 713,867	52 169,161	68,34 %	—
2025-03-28	111 338,000	28 725,204	47 364,670	0,782	0,858	36 732,997	55 203,578	68,34 %	—
2025-03-29	99 141,000	25 578,378	42 175,903	0,782	0,858	32 708,923	49 156,064	68,34 %	—
Total	2 886 397,000	744 690,426	1 227 911,768	—	—	952 289,547	1 431 132,598	68,34 %	

Non-production days (7): 2025-03-02, 2025-03-09, 2025-03-16, 2025-03-23, 2025-03-24, 2025-03-30, 2025-03-31. These dates have no row in daily_production and are reported as N/A by omission. Weekends and public holidays in Colombia during March 2025 account for all of them.

Methodology & References

Volumetric conversion applies the canonical relation `litres = kg / density` at the calibrated product density measured at 15 °C for the reporting month. Densities are resolved per production month from the platform `product_densities` table (one row per (product, effective_from) pair), so historical conversions remain reproducible if downstream batches recalibrate the product specification. The densities applied to March 2025 are:

- **EU-grade (DEV-P100-EU)** — $\rho_{\text{EU}} = 0,782 \text{ kg/L}$
- **PLUS-grade (DEV-P100-PLUS)** — $\rho_{\text{PLUS}} = 0,858 \text{ kg/L}$

Both density values are ISCC-recognised product-specification figures for the OisteBio DEV-P100 — refined pyrolysis oil family. They are seeded and maintained in the platform `product_densities` table — initially loaded by Alembic migration `0005_seed_product_densities.py` and extended with per-month effective rows by `0014_monthly_densities.py`. Once a reporting period closes, the row applicable to that month becomes immutable for audit purposes; new effective rows are appended forward, never back-dated.

Per-row litre values shown in the daily table are not stored independently — they are derived deterministically inside PostgreSQL as **GENERATED ALWAYS** columns on the `daily_production` table:

```
litres_eu    NUMERIC GENERATED ALWAYS AS (eu_prod_kg / 0.782) STORED,
litres_plus NUMERIC GENERATED ALWAYS AS (plus_prod_kg / 0.858) STORED
```

These columns are introduced by Alembic migration `0007_persist_production_litres.py`. Once persisted, historical litre values become immutable for audit purposes even if product densities are revised in the future. Write access to `litres_eu` and `litres_plus` is prohibited at the schema level: any attempt to update them raises PostgreSQL error code 42601.

The **input kg** column reports the daily value of `daily_production.kg_to_production` — the mass of pre-conditioned ELT feedstock routed to the pyrolysis reactor that calendar day. The **yield %** column reports the fraction of input mass that exits the refining line as liquid product (EU + PLUS), with the remainder distributed across carbon black, steel scrap, water condensate, syngas, and recorded losses (totals reported in the Executive Summary above and reconciled in Annex A).

This Production Conversion Log accompanies Annex A (mass balance) and Annex D (closing-stock carry-over) in the RTFO-310825 submission bundle and is cryptographically anchored to the cover letter via the SHA-256 digest reported in the running footer (short prefix `000000000000`). The authoritative digest is reproduced in `MANIFEST.sha256` at the bundle root.

— End of Production Conversion Log —